

HEMPHILL COUNTY AP, TX, USA

WMO: 722637

Lat: 35.900N

Lon: 100.400W

Elev: 2396

StdP: 13.47

Time Zone: -6.00 (NAC)

Period: 03-19

WBAN: 93046

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.6	16.5	-2.0	5.4	27.3	2.7	6.9	34.7	28.6	50.7	26.1	52.9	5.8	20	0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.4	100.4	68.9	98.6	68.8	96.0	69.2	75.4	89.9	74.3	88.9	73.2	87.6	13.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.8	128.6	80.6	70.4	122.6	79.5	69.4	118.3	78.9	40.6	90.0	39.5	89.2	38.5	88.0	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.0	24.5	21.4	DB	3.0	105.1	7.6	4.1	-2.5	108.0	-7.0	110.5	-11.3	112.8	-16.8	115.8	(4)
(5)			WB	2.1	77.2	7.4	2.5	-3.2	79.0	-7.6	80.4	-11.7	81.8	-17.1	83.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.8	38.4	40.3	51.6	59.0	67.9	78.1	81.3	79.4	72.5	60.0	48.5	39.2		
	DBStd	17.67	9.97	11.11	9.95	9.34	8.74	5.93	5.30	5.62	7.14	9.14	9.45	9.24		
	HDD50	1319	379	303	100	27	2	0	0	0	0	21	138	349		
	HDD65	3768	825	692	427	225	73	0	0	0	0	208	500	800		
	CDD50	4886	19	32	150	296	556	842	970	911	675	329	92	14		
	CDD65	1861	0	1	12	44	162	392	506	446	243	51	4	0		
	CDH74	23706	11	57	254	756	2086	4862	6629	5368	2772	775	131	5		
	CDH80	12220	0	16	72	295	959	2554	3803	2945	1273	272	31	0		
	WSAvg	8.9	7.4	8.5	9.9	11.3	10.1	10.5	8.6	7.8	8.6	8.3	8.2	7.8		
	PrecAvg	24.3	0.8	0.7	1.5	2.2	3.5	3.5	2.6	2.9	2.0	2.2	1.1	1.0		
(15) Precipitation	PrecMax	35.5	2.0	2.3	4.9	10.9	7.8	9.1	6.5	6.3	5.0	8.1	4.2	3.9		
	PrecMin	17.8	0.0	0.0	0.0	0.0	0.1	0.2	0.6	0.2	0.0	0.1	0.0	0.1		
	PrecStd	4.0	0.6	0.7	1.3	2.2	1.6	2.0	1.4	1.3	1.3	2.0	1.1	1.0		
	PrecStd	4.0	0.6	0.7	1.3	2.2	1.6	2.0	1.4	1.3	1.3	2.0	1.1	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.1	81.0	87.1	93.6	98.7	104.3	105.7	103.9	98.8	91.6	84.1	72.9		
		MCWB	50.1	51.8	56.5	57.0	62.0	65.7	67.7	70.1	67.0	62.4	55.2	49.4		
	2%	DB	68.7	73.2	80.9	88.3	94.6	99.0	100.6	99.7	95.1	87.5	78.0	66.5		
		MCWB	47.8	50.4	56.6	57.1	63.5	67.2	69.2	69.7	68.4	63.3	54.3	48.2		
	5%	DB	62.6	66.5	75.3	82.1	90.2	95.4	98.7	97.2	91.3	82.0	72.0	61.1		
		MCWB	45.1	48.5	53.7	58.0	63.5	69.2	69.7	70.3	68.4	61.6	53.2	45.4		
	10%	DB	56.7	60.1	70.4	77.3	85.7	92.0	95.8	94.2	87.9	77.7	66.4	55.8		
		MCWB	42.7	45.1	52.4	57.0	63.8	70.0	70.5	70.7	67.5	60.3	51.8	43.9		
	0.4%	WB	52.8	57.0	64.3	67.3	73.4	76.6	77.4	76.5	74.6	71.8	63.1	55.9		
		MCDB	69.3	70.5	74.6	78.8	84.8	90.6	91.2	90.1	87.2	80.7	68.2	61.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	49.5	52.2	62.0	64.8	71.1	74.9	75.5	74.8	72.8	68.3	58.3	50.8		
		MCDB	64.4	67.8	71.2	74.3	81.6	89.3	90.4	89.7	85.5	77.9	67.2	62.8		
	5%	WB	46.4	49.6	59.0	62.2	69.2	73.6	74.4	73.5	71.3	66.1	55.6	47.4		
		MCDB	60.9	65.8	68.4	72.2	80.5	88.0	89.2	88.9	83.5	75.1	67.7	59.5		
	10%	WB	43.6	46.4	55.0	59.8	67.3	72.4	73.1	72.4	70.0	63.4	53.0	44.3		
		MCDB	55.4	59.5	67.2	71.5	79.4	86.2	87.6	86.9	81.7	73.1	65.4	55.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	27.5	26.5	27.9	27.1	25.5	24.2	24.4	24.5	25.2	27.7	28.7	26.0		
		MCDBR	37.0	38.8	34.8	35.3	31.9	28.3	28.2	28.9	28.6	34.4	37.3	35.1		
		MCWBR	19.5	19.6	15.8	14.5	11.8	8.7	6.9	7.6	9.2	14.6	18.0	19.6		
		MCDBR	34.2	35.7	27.1	25.4	25.6	24.0	24.2	24.4	23.0	25.0	29.5	30.8		
	5% WB	MCWBR	18.9	19.7	14.2	13.6	11.9	9.7	8.4	8.1	9.0	13.2	16.5	18.2		
		MCWBR	18.9	19.7	14.2	13.6	11.9	9.7	8.4	8.1	9.0	13.2	16.5	18.2		
(40) Clear-Sky Solar Irradiance	taub		0.267	0.274	0.309	0.339	0.356	0.392	0.395	0.384	0.351	0.318	0.286	0.269		
	taud		2.565	2.555	2.464	2.402	2.424	2.365	2.391	2.383	2.456	2.526	2.552	2.567		
	Ebn at Noon		301	311	305	299	294	282	280	282	287	290	291	294		
	Edh at Noon		24	27	33	37	37	39	38	37	33	28	24	23		
(44) All-Sky Solar Radiation	RadAvg		942	1184	1563	1933	2148	2318	2271	2028	1725	1289	998	816		
	RadStd		84	125	114	127	183	162	105	107	86	162	104	65		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+196	N/A

Nomenclature: See separate page